

2024 Calculus II Additional Exercise (11)

1. Exercise 1, 2, 3, 6(first), 8, 13 in page 46 from Calculus Supplement Homework.
2. Exercise 17, (6), (7), (14)($\iiint$ is revised as $\iint$), (16) in page 47 from Calculus Supplement Homework.
3. Exercise 40 in page 50 from Calculus Supplement Homework.
4. Find

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} \, dy \, dx.$$

5. Evaluate $\iiint_D \frac{x^2}{a^2} \, dV$, where $D : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ is an ellipsoid.